SECTIONS WITH SUPPORT PROPER OVER THE BASE FORM AN ABELIAN SUBSHEAF OF THE PUSHFORWARD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Support of a section is closed (Etale Cohomology, Lemma etale-cohomology-lemma-support-section-closed, canonical tag 04FT). cited at proof lines 394-395. Closes the support, enabling the proper-over-base material to apply.
- Proper over base material (Cohomology of Schemes, Section coherent-section-proper-over-base, canonical tag 0CYK). cited at proof lines 396-397. The body of results the argument builds on.
- Unions of closed subsets proper over the base (Cohomology of Schemes, Lemma coherent-lemma-union-closed-proper-over-base, canonical tag 0CYR). cited at proof lines 398-400. Makes the subset of f_*F(V) a subgroup.
- Base change closedness proper over the base (Cohomology of Schemes, Lemma coherent-lemma-base-change-closed-proper-over-base, canonical tag 0CYP). cited at proof lines 401-403. Makes the rule a sub presheaf.
- Stalks of pullback sheaves (Etale Cohomology, Lemma etale-cohomology-lemma-stalk-pullback, canonical tag 03Q1). cited at proof lines 408-410. Together with the stalk characterization of the support it identifies the support of s|_{V_i} as the inverse image of the support of s.
- Descending property of properness over the base (Descent, Lemma descent-lemma-descending-property-proper-over-base, canonical tag 0F4J). cited at proof lines 411-412. Descends properness of the support from the etale covering to V, giving the sheaf condition.

Source reference. Original lemma 04FT: sources/stacks-chow/tags.txt, source lines 5809–5809. Cited at proof lines 394-395: the support of a section is closed, so the proper-over-base material applies.

Source reference. Original section 0CYK: sources/stacks-chow/tags.txt, source lines 16245–16245. Cited at proof lines 396-397: the body of results on supports proper over the base that the argument builds on.

Source reference. Original lemma 0CYR: sources/stacks-chow/tags.txt, source lines 16251–16251. Cited at proof lines 398-400: makes the subset of f_*F(V) a subgroup.

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Source reference. Original lemma 0CYP: sources/stacks-chow/tags.txt, source lines 16249–16249. Cited at proof lines 401-403: makes the rule a sub presheaf of f_*F.

Source reference. Original lemma 03Q1: sources/stacks-chow/tags.txt, source lines 4878–4878. Cited at proof lines 408-410: with the stalk characterization of the support it identifies the support of s|_{V_i} as the inverse image of the support of s.

Source reference. Original lemma 0F4J: sources/stacks-chow/tags.txt, source lines 18904–18904. Cited at proof lines 411-412: descends properness of the support from the etale covering to V, closing the sheaf condition.

Lemma 0.2. *Let $f : X \to Y$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$ be an abelian sheaf on $X_{étale}$. The rule*

$$Y_{étale} \longrightarrow Ab, \quad V \longmapsto \{s \in f_*\mathcal{F}(V) = \mathcal{F}(X_V) \mid Supp(s) \subset X_V \text{ is proper over } V\}$$

is an abelian subsheaf of $f_*\mathcal{F}$.

Proof. Recall that the support of a section is closed (Étale Cohomology, Lemma 04FT) hence the material in Cohomology of Schemes, Section 0CYK applies. By the lemma above and Cohomology of Schemes, Lemma 0CYR we find that our subset of $f_*\mathcal{F}(V)$ is a subgroup. By Cohomology of Schemes, Lemma 0CYP we see that our rule defines a sub presheaf. Finally, suppose that we have $s \in f_*\mathcal{F}(V)$ and an étale covering $\{V_i \to V\}$ such that $s|_{V_i}$ has support proper over V_i. Observe that the support of $s|_{V_i}$ is the inverse image of the support of $s|_V$ (use the characterization of the support in terms of stalks and Étale Cohomology, Lemma 03Q1). Whence the support of s is proper over V by Descent, Lemma 0F4J. This proves that our rule satisfies the sheaf condition. □